PHASE 4 DOCUMENT

EduGenie: Google Gemini Powered Learning Assistant

Phase Name: AI Learning Assistant Interface and Core Learning Features

Project Name	EduGenie: Google Gemini Powered Learning Assistant
Phase	Phase 4 – AI Interface and Core Learning Features
Project Type	AI-Powered Learning Application
Main Technology	Google Gemini API
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 4 of **EduGenie: Google Gemini Powered Learning Assistant**. After establishing the secure Google Gemini backend connection in Phase 3, this phase focuses on the student-facing interface and the core learning experience. It connects the EduGenie dashboard, AI chat, explanation controls, response display, and saved-learning features with the completed backend service.

2. Phase Objective

The objective of this phase is to create a simple, responsive, and student-friendly interface through which learners can ask questions, choose a learning mode, receive AI-generated explanations, continue the conversation, and save useful material. The interface should make AI support easy to access without making the learning process confusing or overwhelming.

3. Student Dashboard

The student dashboard is the main starting point after login. It provides quick access to the core EduGenie features: Ask EduGenie, Explain a Topic, Summarize Notes, Generate Quiz, Create Study Plan, and Project Help. The dashboard can also show recent sessions, saved responses, suggested prompts, and reminders for responsible use of AI-generated content.

4. AI Chat Interface

The AI chat screen includes a text-input area where students can ask a question in natural language. It displays student messages and EduGenie responses in a readable conversation layout. The interface includes a send action, loading indicator, retry option for failed requests, and follow-up support. Students can ask questions such as “Explain recursion with a simple example” or “Help me understand the difference between a class and an object.”

5. Learning Mode Selection

Before sending a request, the student can select a learning mode. Suggested modes include Explain Simply, Explain in Detail, Give Example, Step-by-Step Help, Summarize, Create Practice Questions, and Project Guidance. The selected mode is sent to the backend so it can construct an appropriate Gemini prompt. This gives students more control over the type and depth of assistance they receive.

6. Response Presentation

EduGenie should present AI responses in a structured and readable way. The interface may display a short answer first, followed by explanations, examples, key points, and a suggested next step. Basic formatting such as headings, bullet points, numbered steps, and code blocks may be supported where applicable. Responses should include a reminder that important academic information should be checked with faculty, textbooks, and approved resources.

7. Conversation Continuity

Students may need to ask follow-up questions after receiving an answer. The application should allow a user to continue the same learning session so that the next request can refer to the previous topic. The backend can use limited, relevant session context to improve continuity while controlling context size and avoiding unnecessary storage of sensitive information.

8. Save and Review Learning Content

A student can save helpful responses, explanations, summaries, or examples for later revision. Saved items can be shown in a My Learning or Saved Content section, organized by subject, topic, date, or learning type. Students should be able to view and delete their own saved items. This feature turns one-time AI responses into reusable learning resources.

9. Input and Output Controls

The interface should prevent empty submissions, clearly show a loading state while Gemini is responding, and display friendly error messages when a request fails. It should limit excessively long input if required and offer suggestions for writing clearer questions. Output controls can include copy, save, ask a follow-up question, generate an example, or request a simpler explanation.

10. Accessibility and Responsive Design

EduGenie should work well on mobile phones, tablets, and desktop browsers. The user interface should use readable font sizes, adequate contrast, clear labels, keyboard-friendly controls where practical, and responsive layouts. Buttons should be easy to identify and touch on a mobile device. The application should avoid depending only on color to convey meaning and should provide understandable messages for all states.

11. Responsible Learning Guidance

The interface should state that EduGenie is an educational support tool and that AI content can contain errors or incomplete information. Students should be encouraged to use the tool for understanding and practice, not for copying answers without learning. The platform can include a notice such as: “Use this explanation to learn; verify important answers with your faculty or prescribed material.”

12. Functional Workflow

1. Student logs in to EduGenie.
2. Student opens the dashboard and selects Ask EduGenie or another learning option.
3. Student selects a learning mode.
4. Student enters a topic or question.
5. The frontend validates the input and shows a loading state.
6. The request is sent to the secure Gemini-enabled backend.
7. The response is received and displayed in the chat interface.
8. Student reads the explanation, asks a follow-up, copies it, or saves it.
9. Saved learning content is available in the student’s profile area.

13. Testing Plan

Testing should confirm that the dashboard links open the correct features, the chat input accepts valid questions, empty input is blocked, learning-mode selection is sent correctly, loading indicators appear, responses display properly, follow-up questions work, saved content is available only to the relevant user, and error messages are understandable. Responsive testing should be performed on common mobile and desktop screen sizes.

14. Phase Deliverables

The deliverables of this phase are the student dashboard, AI chat interface, learning-mode selector, response-display component, conversation-continuity logic, save-and-review content feature, input and output controls, responsive UI design, responsible-use notice, and test results for core user interactions.

15. Planned Next Phase

The next phase will be **Phase 5: Quiz Generation, Study Planner, and Learning Progress Features**. It will extend the core AI chat experience by adding structured practice quizzes, revision planning, quiz attempts, progress tracking, and personalized learning-support tools.

16. Conclusion

Phase 4 brings the EduGenie learning experience to the student interface. By connecting a clear dashboard and chat experience to the secure Gemini backend, the system enables students to ask questions, receive tailored explanations, continue learning conversations, and save useful resources. This phase provides the usable foundation for the advanced learning tools planned next.

Phase Approval

Prepared by	_____
Guide / Faculty	_____
Approved by	_____